# BIA575: Data Storage Methods

POST UNIVERSITY - WATERBURY CAMPUS
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Post UNIVERSITY®
THE MALCOLM BALDRIGE SCHOOL OF BUSINESS®

## Course Information

BIA575: Data Storage Methods
Credits: 3 Semester Hours

Course Description

This course provides an in-depth exploration of data warehouses and contemporary data storage methodologies employed in today's organizations. We will delve into various types of data storage and databases, examining their security, encryption, and operational dynamics, while also considering their strategic alignment within the broader organizational context. Students will gain hands-on experience with both PowerBI, one of the leading platforms in analytic dashboarding and Structured Query Language (SQL).

## Course Materials

**Required:**

- Gillenson, M. (2023). *Fundamentals of Database Management Systems* (3rd Edition). Wiley.

## BIA Program Outcomes

Students in the BIA Program will:

1. Assess the organization's business for business intelligence and analytics systems.
2. Formulate business intelligence and analytics systems and process requirements.
3. Organize business intelligence and analytics team and responsibility requirements.
4. Correlate business and processes related to business intelligence and analytics applications

## Course Outcomes

Students participating in this course will:

1. Evaluate data storage architectures and their uses.
2. Synthesize strategic data management plans.
3. Generate innovative data management techniques.
4. Design advanced analytic dashboards.

## Course Procedures

This course is organized into a series of weekly lessons (Units) composed of readings, discussions among and between the students and instructor, and “homework” assignments, which build upon the readings and discussion. Each Unit has a closing date of Sunday at 11:59 p.m. unless otherwise stated). *You must have participated in the discussion and handed in the assignment by the end of the Unit date.*
It is expected that the materials provided in the units will be reviewed as well as any additional materials, notes or presentations by the instructor.

The course is laid out with the following process in mind:

1. Review the Readings assigned for a topic

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2. Participate in the Discussion – ask questions
3. Complete the Homework Assignment – per the syllabus
4. Set aside time each week for the Projects and review them quickly so you do not wait until the last moment.
5. Each week's assignment builds toward the final project, where you will apply the course concepts and the program concepts of innovation and practical application of concepts.

## Grading

Grades for this course will be based upon

- Discussions - 25%
- Assignments - 55%
- Journals - 20%

### Discussion Board Participation and Grading

Success in the BIA Program depends in very large part on the student's participation in the class discussions (Discussion Board). The Student is accountable for understanding and complying with the general standards for participation in Class Discussions, the requirements in the BIA Discussion Board Rubric, and the individual instructor's guidelines.

*See the item in **Course Information** titled **Discussion Board Criteria and Grading Rubric** for more specific expectations of quality participation and the Grading Rubric.*

### Assignment Grading

Each assignment has an assignment number and due date. Assignments are due on the Sunday of the week they are assigned by midnight EST unless otherwise noted. Assignments are to be submitted on time and throughout the course.

***See the assignment area under each unit for assignment criteria and grading rubric for more specific expectations.***

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## Course Outline

| Unit | Topic | Assignment |
|---|---|---|
| 1 | Introduction to Data Storage and Management | • Unit 1 Discussion Board: Data as a Competitive Advantage<br>• Unit 1 Assignment: Entity-relationship Diagram |
| 2 | Database Management Systems Concepts, SQL, and the Relational Database Model | • Unit 2 Assignment: SQL Essential Training<br>• Unit 2 Assignment: SQL Journal |
| 3 | Logical and Physical Database Design, Object-Oriented Database Management | • Unit 3 Discussion Board: Database Design Principles PowerPoint<br>• Unit 3 Assignment: Relational Databases Essential Training |
| 4 | Database Management, Administration, and PowerBI | • Unit 4 Discussion Board: Database Administration in Your Organization<br>• Unit 4 Assignment: Power BI Essential Training<br>• Unit 4 Assignment: PowerBI Dashboard Creation |
| 5 | Database Control, Security, and Data Warehousing | • Unit 5 Discussion Board: Data Security in Action |
| 6 | Data Visualization and Analytics, Database Applications | • Unit 6 Discussion Board: Advanced Data Visualization Techniques<br>• Unit 6 Assignment: Journal – Summarize Your Learning<br>• Unit 6 Assignment: Capstone Contract |
| 7 | AI Data Models and Database Architecture | • Unit 7 Discussion Board: AI in Database Architecture<br>• Unit 7 Assignment: GenAI and Predictive AI Architecture Foundations<br>• Unit 7 Assignment: Introduction to Prompt Engineering for Generative AI (2023)<br>• Unit 7 Assignment: The Future of AI in Data Storage |
| 8 | Emerging Trends and Professional Networking | • Unit 8 Discussion Board: Professional Networking<br>• Unit 8 Discussion Board: Resume<br>• Unit 8 Assignment: Emerging Topics of Interest |

**Post University Credit Hour Expectations**

Post University takes the integrity of our coursework very seriously. A credit hour is an academic unit earned for engaged academic time, including classroom or online instruction and outside study. This expectation varies based on the type of class in which you are enrolled.

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**Main Campus Courses**
For a 3-credit semester course, you are expected to engage in three hours of in-class learning and six hours of outside study for a total of nine hours of engaged academic time per week for 15 weeks.

**Accelerated Online or Hybrid Site Courses**
For an accelerated 8-week, 3-credit hybrid site or online course, you are expected to engage in 17 hours of engaged academic time per week; for a hybrid site course, 3 hours will be on site.

**Blended Courses**
Blended courses vary in the balance of classroom and online instruction. For a 3-credit blended course, students are expected to complete a total of 135 hours of engaged academic time, whether it is a semester-long or accelerated format.

**What is 'engaged academic time'?**
This is when you are engaging in academic learning activities related to a course. These activities include both in-class, out-of-class, and online activities such as readings, lectures, group work, videos, presentations, webinars, research, writing, surveys, discussions, practice exercises, web development, creating, producing, studying, and any other learning experiences that contribute to the achievement of course outcomes.

Your responsibility as a student at Post University is to fulfill these academic expectations in the time you devote to this class.

***To ensure that we continue to meet our academic standards and your learning expectations, we routinely assess our programs, courses, and instructors. Completion of the end-of-course Student Survey is a required component of this course.***

***For the full Academic Credit Hour Policy, please refer to the Post University Catalog and/or the Post University website.***

## University Policies and Procedures*

Following are the policies and procedures for graduate courses at Post University. These are too considered part of the course syllabus for the course for which I am the instructor. Please read this document carefully. If you have any questions about the contents of this document, please ask.

**Email**
When using email correspondence, please use the class identifier in the Subject area. For example: BUS510 and sign your message with your name so the instructor knows who it is from.

**Scheduled Absences**
If you will miss class for religious reasons, work obligations, military duty, or have other absences you are aware of, please inform the instructor in writing via email at least two weeks prior to the date you will be absent from class.

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**Unearned F Policy**
Post University recognizes that student participation in courses is essential to student success. Course participation is assessed by faculty at the time that final course grades are determined. Faculty members will assign an unearned F grade to a student who does not complete graded academic activity after the midpoint of a course. When a student receives an unearned F grade, the faculty member will determine the date the student submitted his or her last graded academic activity. This will be considered the student's withdrawal date from the course.

As a result of receiving an unearned F, a student may be administratively withdrawn. The University may be required to calculate how much financial aid the student has earned for the term. This calculation may result in the return of unearned financial aid to the U.S. Department of Education or other third parties. The student may be obligated to pay for charges originally paid by the unearned financial aid and/or return any excess funds paid to the student. Receiving an unearned F may also impair future course registrations and financial aid availability. Unearned F grades also count as failing grades in the student's GPA and pace of completion. Please see the policies on Administrative Withdrawal and Satisfactory Academic Progress.

**Academic Dishonesty**
Academic dishonesty includes but is not limited to, willful or inadvertent plagiarism, cheating, unauthorized collaboration, unauthorized use of digital materials, the introduction of viruses, willful interruption of systems, and may also include unintentional complicity in other's academic dishonesty.

**Professional Conduct and Civility**
Post University's graduate programs assume that all students attend for a serious purpose. Students are expected to conduct themselves as responsible individuals with high standards of academic integrity. The University will not condone academic dishonesty in any form nor tolerate violations of professional ethics within the academic community. Post University reserves the right to dismiss a student from the graduate school for any behavior deemed to be unprofessional or unethical.

***Note: All and full University policies can be found in the University Catalog, Graduate Studies Handbook, University Policies link in your course, and/or Website.***

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